

Importing a Single Class

Let's create a module containing just the `Car` class. This brings up a subtle naming issue: we already have a file named `car.py` in this chapter, but this module should be named `car.py` because it contains code representing a car. We'll resolve this naming issue by storing the `Car` class in a module named `car.py`, replacing the `car.py` file we were previously using. From now on, any program that uses this module will need a more specific filename, such as `my_car.py`. Here's `car.py` with just the code from the class `Car`:

```
car.py ❶ """A class that can be used to represent a car."""

class Car:
    """A simple attempt to represent a car."""

    def __init__(self, make, model, year):
        """Initialize attributes to describe a car."""
        self.make = make
        self.model = model
        self.year = year
        self.odometer_reading = 0

    def get_descriptive_name(self):
        """Return a neatly formatted descriptive name."""
        long_name = f"{self.year} {self.make} {self.model}"
        return long_name.title()

    def read_odometer(self):
        """Print a statement showing the car's mileage."""
        print(f"This car has {self.odometer_reading} miles on it.")

    def update_odometer(self, mileage):
        """
        Set the odometer reading to the given value.
        Reject the change if it attempts to roll the odometer back.
        """
        if mileage >= self.odometer_reading:
            self.odometer_reading = mileage
        else:
            print("You can't roll back an odometer!")

    def increment_odometer(self, miles):
        """Add the given amount to the odometer reading."""
        self.odometer_reading += miles
```

We include a module-level docstring that briefly describes the contents of this module ❶. You should write a docstring for each module you create.

Now we make a separate file called `my_car.py`. This file will import the `Car` class and then create an instance from that class:

```
my_car.py ❶ from car import Car

my_new_car = Car('audi', 'a4', 2024)
print(my_new_car.get_descriptive_name())
```